Year 8 SEAL Science — Energy Practice Test (Easy)

TEACHER MARKING GUIDE — not for distribution to students

Total 40 marks: Section A 12 × 1 mark (separate answer sheet), Section B 28 marks. Writing time 55 min. This is a practice test at an easier level: short single-step calculations, formula reminders in the stems and sentence starters on explanations. Marking conventions: one mark per idea or step; no half marks. Mark units once, at the final answer. Where a later part uses an earlier answer, mark it consequentially.

Section A — Multiple-choice answer key (12 marks)

Key balance: A ×3, B ×3, C ×3, D ×3; no letter appears twice in a row. Each wrong letter points to one misconception or slip, so a tally of wrong letters shows what to reteach.

Q	Key	Working for the key	Distractor diagnosis
1	B	Food stores chemical energy, a type of potential energy.	A: food lets you move, so thinks it is kinetic. C: thinks food energy is heat. D: mixes up the two kinds of potential energy.
2	A	GPE = 60 × 10 × 3 = 1,800 J	B: left out *g*. C: left out *h* (*m* × *g*). D: added ½ as if it were KE.
3	C	Radiation travels through empty space from the Sun.	A: thinks heat from the Sun needs particles to carry it. B: thinks hot air comes from the Sun (convection needs a fluid, and hot air rises). D: confuses heating by the Sun with friction.
4	D	Convection: warm air is less dense and rises; cooler, denser sea air flows in.	A: thinks heat or cold 'attracts'. B: 'heat rises' with no density idea. C: thinks waves move the wind.
5	A	Most of the 320 J becomes thermal energy (a little is reflected).	B: energy destroyed. C: thinks a panel stores energy like a battery. D: thinks energy can stay trapped unchanged.
6	B	P 1 ÷ 4 = 25%; Q 2 ÷ 5 = 40%; R 3 ÷ 20 = 15%	A: picks the smallest energy input. C: picks the most light out, not the fraction. D: confuses conservation with efficiency.
7	C	Fire is used to care for Country: new green growth after a fire attracts animals.	A: thinks fire is used to heat soil for farming. B: thinks fire makes energy from nothing (it releases stored chemical energy). D: thinks fire is used to destroy all vegetation.
8	D	More stars = more efficient: less energy for the same job.	A: stars mean power or strength. B: stars show price. C: more powerful means more stars.
9	A	6 × 5 = 30 Wh; 30 ÷ 1,000 = 0.03 kWh	B: Wh not changed to kWh. C: divided power by time. D: divided by 100, not 1,000.
10	B	Foam is an insulator and slows heat flow into the esky.	A: confuses insulator and conductor. C: 'cold' as a substance that flows. D: thinks insulators stop heat completely.
11	D	1,100 Wh needed > 960 Wh available	A: reversed supply and demand. B: didn't compare the numbers. C: thinks a battery makes energy.
12	C	Averaging repeats reduces the effect of random errors.	A: results should fit the hypothesis. B: confuses repeats with variables. D: confuses repeats with instrument precision.

Section B — Short-answer marking scheme (28 marks)

Before marking, show the marking team the rejected answers listed under each part so the team marks the same way.

Question 1 — Surf day (6 marks)

a) (2 marks)

1 mark (i) sound energy, kinetic. **1 mark** (ii) gravitational (potential) energy, potential. Both the type and the family needed for each mark

Reject: (i) 'kinetic, kinetic' scores 0, because the type is sound; (ii) 'stored energy' with no type named

b) (2 marks)

1 mark KE = ½ × 70 × 4² (or ½ × 70 × 16). **1 mark** 560 J

Reject: 280 J (did not square the speed) scores 0; 1,120 J (left out the ½) scores 1 mark only

c) (2 marks)

1 mark height = 4,200 ÷ (70 × 10). **1 mark** 6 m

Reject: 60 m (left out *g*) scores 0 for the answer; a value with no unit loses the second mark

Question 2 — Campfire (4 marks)

a) (2 marks)

1 mark (i) conduction. **1 mark** (ii) convection

Reject: (ii) radiation: the socks are dried mainly by the rising hot air; 'heat rises' alone is not a method

b) (2 marks)

1 mark particles at the hot end gain energy and vibrate faster. **1 mark** they bump into their neighbours and pass the energy along, particle by particle, so the energy moves along the peg (metal is a good conductor)

Reject: 'The heat moves up the peg' with no particles; 'the particles get bigger' or 'the particles melt'; 'the particles move along the peg to the other end' (they vibrate in place in a solid)

Question 3 — Camp stove (3 marks)

a) (2 marks)

1 mark 1,200 ÷ 3,000 × 100. **1 mark** 40%

Reject: 1,200 ÷ 1,800 × 100 = 67% (useful ÷ wasted) scores 0; 60% (wasted fraction) scores 0

b) (1 mark)

1 mark any one link to less waste: the windshield stops the wind carrying hot air away from the flame and pot (less convection); or the metal shield reflects radiation back towards the pot; or less heat is lost to the surroundings, so more goes into the water. Useful output rises for the same gas

Reject: 'It makes more energy'; 'It blocks the wind' with no link to less energy wasted or more energy into the water

Question 4 — Electric cooler (4 marks)

a) (2 marks)

1 mark 50 × 24 × 10 = 12,000 Wh. **1 mark** 12,000 ÷ 1,000 = 12 kWh

Reject: 12,000 kWh (Wh not converted) scores the first mark only

b) (1 mark)

1 mark 12 × $0.30 = $3.60

Consequential: on the kWh value from part a).

Reject: $360 (multiplied by 30 instead of 0.30)

c) (1 mark)

1 mark the electric cooler is cheaper, by $25.00 − $3.60 = $21.40

Consequential: on the cost from part b).

Reject: 'The cooler is cheaper' with no amount

Question 5 — Fire on Country (3 marks)

a) (2 marks)

Correct order: saw the wood (1), ember forms (2), ember into tinder bundle (3), blow into flame (4). **1 mark** steps 1 and 2 both correct. **1 mark** steps 3 and 4 both correct

Reject: No mark for a pair if either of its two numbers is wrong

b) (1 mark)

1 mark any one of: fire drill (hand drill), fire plough, percussion (striking stones for sparks)

Reject: Fire saw (given in the stem); matches or a lighter; 'rubbing two sticks together' on its own (this describes the fire saw too, so it does not name a different method). Accept it only with detail that identifies a method, e.g. 'spinning a stick in a notch' (drill) or 'pushing a stick along a groove' (plough)

Question 6 — Hot caravan (2 marks)

Question 6 (2 marks)

1 mark a passive change, such as: paint the roof white or shiny; add an awning or shade over windows; add insulation or reflective foil in the roof and walls; add roof vents or windows on opposite sides. **1 mark** the matching heat-transfer reason: light or shiny surfaces reflect radiation; shade blocks radiation from the Sun; insulation slows conduction (higher R-value); vents let hot air rise and escape by convection while cooler air flows in

Reject: 'Turn on the air conditioner' or 'use a fan' (appliances); 'park in the shade' (not a change to the caravan's design); 'paint it black'; a change with a reason that does not match it, e.g. 'white paint stops conduction'

Question 7 — Painted cans (6 marks)

a) (1 mark)

1 mark the colour (paint or surface) of the can

Reject: 'the cans'; 'the temperature' (dependent variable); 'the sunlight'

b) (1 mark)

1 mark all three parts, e.g. 'If the can is painted matt black, then its water temperature will rise the most, because dark, dull surfaces absorb more radiation.' Accept a prediction for silver or white rising least because shiny or light surfaces reflect radiation

Reject: No 'because' reason; 'black attracts heat'; a hypothesis that does not name the colour and the temperature rise

c) (1 mark)

1 mark 5.5 °C (shiny silver, trial 2), because it is much higher than the other two silver trials (1.5 and 2.5 °C)

Reject: 'It doesn't fit the pattern' with no values compared

d) (1 mark)

1 mark (1.5 + 2.5) ÷ 2 = 2.0 °C

Consequential: on the outlier chosen in part c).

Reject: 3.2 °C (the outlier included)

e) (2 marks)

1 mark an improvement, such as: do more trials; put a lid on each can; use a more precise (digital) thermometer; keep the thermometer bulb in the middle of the water, not touching the can. **1 mark** its effect: more trials reduce the effect of random errors such as the 5.5 °C result, so the averages are more reliable; lids reduce evaporation, which takes heat away and could differ between cans, so only the colour changes the result (more valid); a digital thermometer reads to 0.1 °C, so small differences between cans show more precisely

Reject: 'Be more careful'; 'use more cans' with no reason; 'test at the same time each day' (the three cans were already tested side by side at the same time); a lid with no reason; an improvement with no effect on the results scores 1 mark only

Mark summary

Section	Marks	Notes
A — multiple choice	12	Answer sheet; key A3 / B3 / C3 / D3
B Q1 — Surf day	6	a 2, b 2, c 2
B Q2 — Campfire	4	a 2, b 2
B Q3 — Camp stove	3	a 2, b 1
B Q4 — Electric cooler	4	a 2, b 1, c 1
B Q5 — Fire on Country	3	a 2, b 1
B Q6 — Hot caravan	2	one part
B Q7 — Painted cans	6	a 1, b 1, c 1, d 1, e 2
Total	40	

Test blueprint

This practice test is pitched easier than the SEAL target: more recall and routine calculation, with sentence starters and formula reminders. It still has a data-based question (Q3, Q7, MC 6), an inquiry-skills question (Q7) and explain parts (Q2b, Q3b, Q6).

Demand level	Marks	Share	Easy target	Items
L1 Recall	11	28%	about 30%	MC 1, 3, 7, 8, 12; SB 1a, 2a, 5b, 7a
L2 Routine procedure	17	42%	about 40%	MC 2, 6, 9, 11; SB 1b, 1c, 3a, 4a, 4b, 5a, 7b, 7d
L3 Interpret or explain	10	25%	about 25%	MC 4, 5, 10; SB 2b, 3b, 4c, 6, 7c
L4 Evaluate	2	5%	about 5%	SB 7e

Coverage by unit lesson

Unit lesson	Marks	Items
2 Forms of energy, KE and GPE	8	MC 1, 2; SB Q1
3 Heat transfer	6	MC 3, 4; SB Q2
4 Transfers, transformations, efficiency, Sankey diagrams	5	MC 5, 6; SB Q3
5 First Nations fire knowledge	4	MC 7; SB Q5
6 Appliances, star ratings, running cost	6	MC 8, 9; SB Q4
7 Household energy design, insulation	3	MC 10; SB Q6
8 Investigation skills	7	MC 12; SB Q7
9 Energy audit: supply and demand	1	MC 11
Total	40	

The Q7 data set has one planted outlier among repeated trials (shiny silver, trial 2). 55 minutes for 40 marks gives about 1.4 minutes a mark; most students should finish with time to check.